Filling the grid

A container of fixed width and height $w, h \in \mathbb{R}^+$ is filled with $n \in \mathbb{N}$ (natural numbers without 0) squares positioned in a grid-like fashion. The squares should cover as much of the $w \times h$ area as possible without overflowing, which means we must maximize the side length $S \in \mathbb{R}^+$ of the square. In other words, we are searching an $b \times a \in \mathbb{N}^2$ grid which fits all our n squares and itself fits in the container while either being as wide or as tall as the container.

We can state the problem in a formal manner as follows:

$$\begin{gathered}
\mathrm{Sol}_a := \left\{ s \in \mathbb{R}^+ \;\middle|\; \exists a, b \in \mathbb{N} \,.\, n \le ab \land as = w \land bs \le h \right\} \\
\mathrm{Sol}_b := \left\{ s \in \mathbb{R}^+ \;\middle|\; \exists a, b \in \mathbb{N} \,.\, n \le ab \land bs = h \land as \le w \right\} \\
\mathrm{Sol} := \mathrm{Sol}_a \cup \mathrm{Sol}_b \\
S := \max \mathrm{Sol}
\end{gathered}$$

Let's start solving for a solution $s_a \in \mathrm{Sol}$ by attempting to find a grid as wide as the container. For any $s \in \mathrm{Sol}_a$:

$$\begin{aligned}
n \le ab \land as = w \land bs \le h &\implies n \le \frac{w}{s} b \land bs \le h \\
&\implies n \le \frac{w}{s} \cdot \frac{h}{s} \\
&\implies n \le \frac{wh}{s^2} \\
&\implies s \le \sqrt{\frac{wh}{n}} \qquad (1)
\end{aligned}$$

This is an upper bound on the side length. Let's determine the grid's number of columns, a:

$$\begin{aligned}
as = w &\implies a = \frac{w}{s} \\
&\overset{(1)}{\implies} a \ge \frac{w}{\sqrt{\frac{wh}{n}}} \\
&\implies a \ge \sqrt{\frac{w}{h} n} \\
(s \text{ maximized} \land a \in \mathbb{N}) &\implies a = \left\lceil \sqrt{\frac{w}{h} n} \right\rceil \qquad (2)
\end{aligned}$$

To further clarify the last step: since s is in the denominator, maximizing it means a must take the smallest value greater than or equal to RHS. Since $a \in \mathbb{N}$ this is precisely RHS rounded up.

Knowing a we can now determine the side length s:

$$as = w \implies s = \frac{w}{a} \tag{3}$$

Since the implications in (1) go in one direction only, we must check that $s \in \mathrm{Sol}$. The first necessary condition is:

$$n \leq \left\lfloor \frac{w}{s} \right\rfloor \cdot \left\lfloor \frac{h}{s} \right\rfloor \tag{4}$$

since $a = \lfloor w/s \rfloor$ and $b = \lfloor h/s \rfloor$ provide the number of squares the grid *should* have on each dimension, if it fits the squares and doesn't overflow. Moreover, $as = w \implies a = w/s = \lfloor w/s \rfloor$, since $a \in \mathbb{N}$.

Let's go to our initial assumptions.

$$n \leq ab \wedge bs \leq h \implies n \leq a\frac{h}{s} \implies \underbrace{n \leq a\left\lfloor \frac{h}{s} \right\rfloor}_{\text{(i)}} \vee \underbrace{(n > a\left\lfloor \frac{h}{s} \right\rfloor \wedge n \leq a\frac{h}{s})}_{\text{(ii)}} \tag{5}$$

In case (i) we can take $b = \lfloor h/s \rfloor$, which means $bs \leq h$. Thus, s as determined above is in $\mathrm{Sol}_a \subseteq \mathrm{Sol}$. But in case (ii), $b = \lfloor h/s \rfloor \implies n > ab \implies s \notin \mathrm{Sol}$. A new solution s' must be found.

Let's analyze what follows from (ii):

$$\begin{aligned} a\left\lfloor \frac{h}{s} \right\rfloor < n \leq a\frac{h}{s} &\implies \left\lfloor \frac{h}{s} \right\rfloor < \frac{h}{s} \wedge n \leq a\frac{h}{s} \\ &\implies \frac{h}{s} < \left\lceil \frac{h}{s} \right\rceil \wedge n \leq a\frac{h}{s} \\ &\implies n < a\left\lceil \frac{h}{s} \right\rceil \end{aligned} \tag{6}$$

This means $b = \lceil h/s \rceil$ is a safe choice for the a that we've determined, and a candidate for s' is:

$$s' = \frac{h}{b} = \frac{h}{\lceil \frac{h}{s} \rceil} \overset{(3)}{=} \frac{h}{\lceil \frac{h}{w}a \rceil} \tag{7}$$

Is this $s' \in \mathrm{Sol}$? Yes, in fact in Sol_b: $(6) \implies n \leq ab$, $(7) \implies bs' = h$ and

$$as' = \frac{ah}{\lceil \frac{h}{w}a \rceil} = \frac{ah}{\lceil \frac{ah}{w} \rceil} \overset{\lceil x \rceil \geq x}{\implies} as' \leq \frac{ah}{\frac{ah}{w}} \implies as' \leq w \tag{8}$$

With that we have determined s_a. Here's its formula based only on inputs w, h, n:

$$a = \left\lceil \sqrt{\frac{w}{h} n} \right\rceil \qquad s_a = \begin{cases} \frac{w}{a} & \text{if } n \leq a \left\lfloor \frac{h}{w} a \right\rfloor \\ \frac{h}{\left\lceil \frac{h}{w} a \right\rceil} & \text{else} \end{cases}$$

Analoguously goes the process for determining s_b, whose formula is:

$$b = \left\lceil \sqrt{\frac{h}{w} n} \right\rceil \qquad s_b = \begin{cases} \frac{h}{b} & \text{if } n \leq b \left\lfloor \frac{w}{h} b \right\rfloor \\ \frac{w}{\left\lceil \frac{w}{h} b \right\rceil} & \text{else} \end{cases}$$

Finally, $S = \max\{\, s_a, s_b \,\}$.

Sadly, this algorithm sometimes fails to find the optimal solution. For example, inputs:

- $w = 10, h = 2, n = 8$ expect $S = 1.25$ but this algorithm gives $S = 1$
- $w = 7, h = 2, n = 20$ expect $S = 0.7$ but this algorithm gives $S = 0.(6)$
- and many others

What do these inputs have in common? When does this algorithm have issues?

To investigate this, let's simplify our solution statement. Observe that:

$$\begin{gathered} s_w \in \mathrm{Sol}_a \iff as_w = w \wedge bs_w \leq h \iff a \geq rb \wedge s_w = \frac{w}{a_w} \\ s_h \in \mathrm{Sol}_b \iff bs_h = h \wedge as_h \leq w \iff rb \geq a \wedge s_h = \frac{h}{b_h} \\ \text{with } n \leq ab \text{ for all of the above} \end{gathered} \tag{9}$$

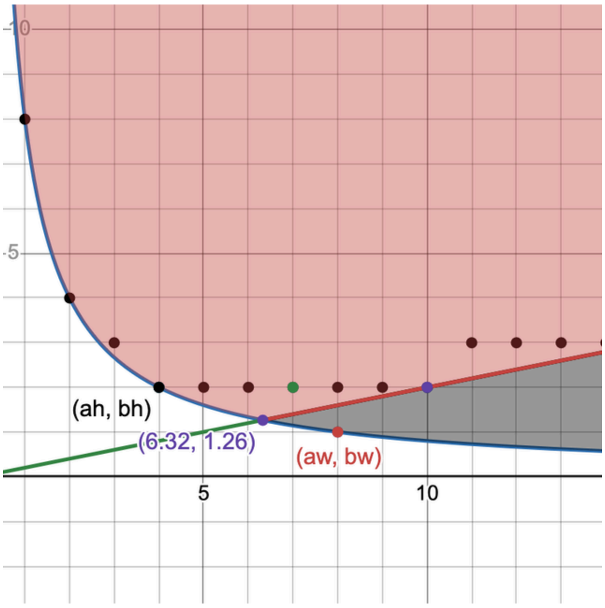
We've simplified width and height to just their ratio, $r = \frac{w}{h}$. Moreover, we don't need the side lengths themselves to compare them. Let $s_w = \frac{w}{a_w}$, $s_h = \frac{h}{b_h}$. Then:

$$s_w > s_h \iff \frac{w}{a_w} > \frac{h}{b_h} \iff \frac{1}{a_w} > \frac{1}{rb_h} \iff a_w < rb_h \tag{10}$$

We can now state the final solution:

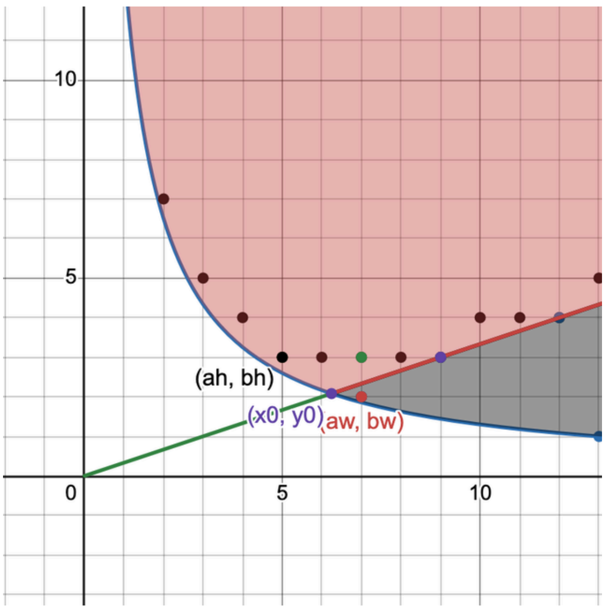
$$\begin{gathered} a_w = \min_{b_w \in \mathbb{N}} \left\{ a \in \mathbb{N} : a \geq \frac{n}{b_w} \wedge a \geq rb_w \right\} \\ b_h = \min_{a_h \in \mathbb{N}} \left\{ b \in \mathbb{N} : b \geq \frac{n}{a_h} \wedge b \geq \frac{a_h}{r} \right\} \\ s = \begin{cases} s_w = \frac{w}{a_w} & \text{if } a_w < rb_h, \\ s_h = \frac{h}{b_h} & \text{otherwise} \end{cases} \end{gathered} \tag{11}$$

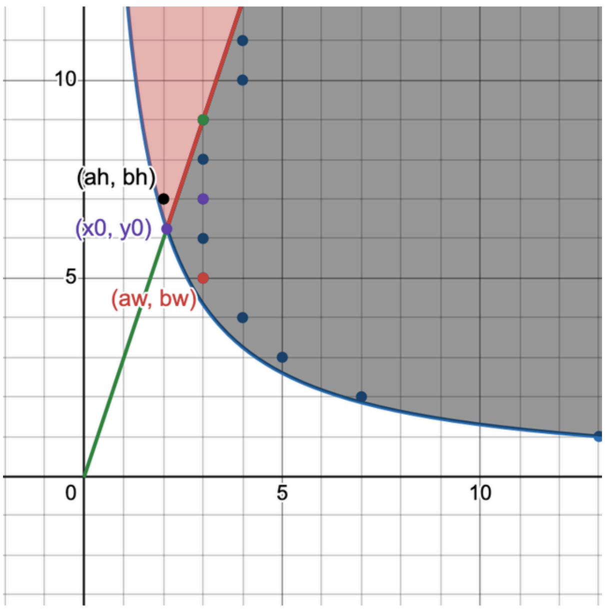
The problem is reduced to just a, b and r. This simplification unlocks a powerful geometric representation in the 2D plane: we're looking for points $(a_w, b_w), (a_h, b_h) \in \mathbb{N}^2$ such that they're on or above the hyperbola $xy = n$ while accounting for their position relative to the line $x = ry$. $x = a$ is used as stand-in for columns in the continuous space, $y = b$ for rows. Take a look (graph A):



This is the graph for the $n = 8, w = 10, h = 2$ example above, with $r = 10/2 = 5$. The black area under $y = rx$ contains all fit-width solutions, the red area all the fit-height solutions. The union of these two contains the entire solution space, all (a, b) for which $n \leq ab$, i.e. all grid dimensions which fit all squares. Points on the green-red line $y = rx$ are grid dimensions having the ratio exactly r. Black points are a subset of other solution candidates.

The symmetry of the problem with respect to ratio can now be directly visualized. For $n = 13$ and $r = 3$, respectively $r = 1/3$, we have:





Terminology-wise, from here on $(a_w, b_w), (a_h, b_h)$ will be called a *fit-width point*, respectively a *fit-height point*, if the points respect the requirements in (9); if the points are those described in (11), they'll also be called *minimal points* (minimized coordinates for maximal side-length). s_w and s_h will be called *fit-width* and *fit-height solutions*; unless explicitly mentioned, the $(a_w, b_w), (a_h, b_h)$ points implied by s_w and s_h are not necessarily minimal. Sometimes only a_w or b_h will be mentioned but keep in mind that by definition they do have a pair b_w and a_h.

Let's now analyze the solution space.

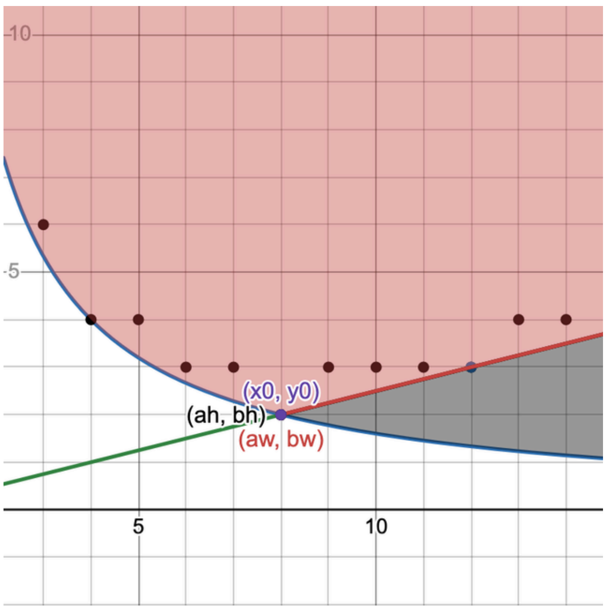
First, notice the purple point at the intersection of the hyperbola and the line – that is, the (x_0, y_0) where $\frac{n}{y_0} = ry_0$ – is special. This would be the "ideal" point, if rows and columns could be fractional somehow: only these dimensions exactly fit all squares and have the exact desired ratio. Notice that:

$$x_0 = \frac{n}{y_0} = ry_0 \iff x_0 = \sqrt{rn} \wedge y_0 = \sqrt{\frac{n}{r}} \tag{12}$$

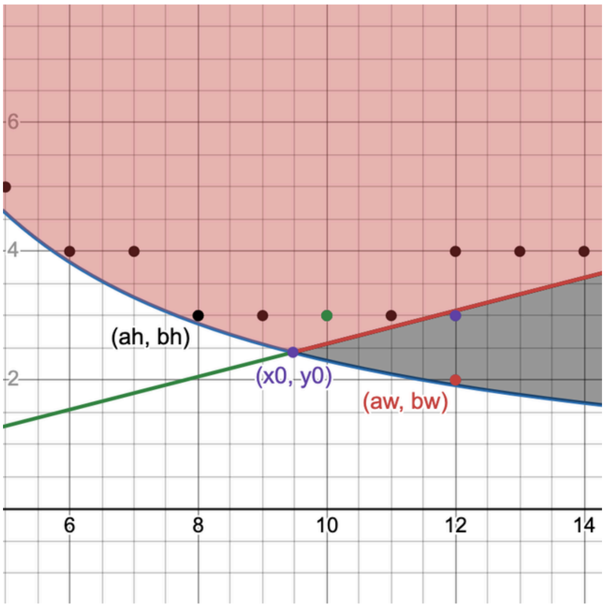
The number of rows a chosen in (2) is precisely $\lceil x_0 \rceil$. This is the visual confirmation of that result: all the fit-width points will find themselves to the left of that point, all the fit-height points to its right. In other words:

$$\forall a_w, b_h \,.\, a_w \geq x_0 \wedge b_h \geq y_0 \tag{13}$$

If parameters are "nice" such that $x_0, y_0 \in \mathbb{N}$ the minimal fit-width and fit-height points coincide (graph B):



Second, when is a point *valid*, i.e. satisfies solution requirements in (9)? Also, can multiple points give the same solution? Consider the following ($n = 23, r = 3.9$, graph C):



The 4 points next to (a_h, b_h) have the same ordinate and the point above (a_w, b_w) the same abscissa, therefore their corresponding side lengths are still s_h, respectively s_w. This follows directly from the solution requirements in (9):

$$\begin{aligned} &\text{Fit width:} \\ a_w b_w \geq n \wedge a_w \geq r b_w &\iff \frac{n}{a_w} \leq b_w \leq \frac{a_w}{r} \\ &\iff \left\lceil \frac{n}{a_w} \right\rceil \leq b_w \leq \left\lfloor \frac{a_w}{r} \right\rfloor \\ &\text{Fit height:} \\ a_h b_h \geq n \wedge b_h \geq \frac{a_h}{r} &\iff \left\lceil \frac{n}{b_h} \right\rceil \leq a_h \leq \lfloor r b_h \rfloor \end{aligned} \tag{14}$$

We can add the ceiling and floor as a consequence of them being residuated mappings. Notice how the intervals describe the space between the hyperbola and the line for a_h and b_w. Doesn't matter which value from these intervals a_h and b_w have, the solutions of (a_w, b_w), (a_h, b_h) are the same. The ceilings and floors are visible in the graph too: the points are a bit above/under the function graphs too. We can compress the inequality to get the validity condition:

$$\begin{aligned} a_w \text{ valid} &\iff \left\lceil \frac{n}{a_w} \right\rceil \leq \left\lfloor \frac{a_w}{r} \right\rfloor \\ b_h \text{ valid} &\iff \left\lceil \frac{n}{b_h} \right\rceil \leq \lfloor r b_h \rfloor \end{aligned} \tag{15}$$

The number of points with equal solutions is easily derived.

Third, which points give the optimal solution s? How do they compare? Let's observe the example graphs for intuition:

- in graph A, a_w is optimal, and $b_w < b_h$
- in graph B, a_w and b_h are equivalent, and $b_w = b_h$
- in graph C, a_w is worse than b_h, and $b_w = b_h$ for one of the fit-width solutions

It would also seem like a fit-width point with more rows than a fit-height point, $b_w > b_h$, gives a worse solution: it's simply a bigger grid in the same space, since the number of columns must also increase ($a_w \geq rb_w$).

Let's prove formally. For $b_w \geq b_h$ we have:

$$b_w \geq b_h \overset{(9)}{\Longrightarrow} a_w \geq rb_w \geq rb_h \geq a_h \implies \frac{1}{a_w} \leq \frac{1}{rb_h} \overset{(10)}{\Longrightarrow} s_h \geq s_w \tag{16}$$

One can read this intuitively as:

> Fit-width solution has at least as many rows and columns, so at least as many squares, thus their side length cannot increase.

This was simple. For $b_w < b_h$ it is a bit more involved:

$$\begin{aligned}
&b_w < b_h \implies rb_w < rb_h \\
&\text{From } (9),\ a_w \geq rb_w. \text{ By total ordering:} \\
&\quad (1) \quad rb_h > a_w \geq rb_w \implies \frac{1}{a_w} > \frac{1}{rb_h} \implies s_w > s_h \\
&\quad (2) \quad a_w \geq rb_h \geq rb_w: \\
&\qquad\quad b_w < b_h \implies \frac{n}{b_w} > \frac{n}{b_h} \implies a_w > \frac{n}{b_h} \\
&\qquad\quad a_w \geq rb_h \wedge a_w > \frac{n}{b_h} \overset{(9)}{\Longrightarrow} (a_w, b_h) \text{ fit-width, same solution} \overset{(16)}{\Longrightarrow} s_h \geq s_w
\end{aligned} \tag{17}$$

Read (17) as:

> 1. Fit-width solution has at most as many columns than the fit-height solution but stretches them out more.
> 2. Fit-width solution can be extended to have as many rows as a fit-height solution but by comparison may not fully stretch them out.

Note that the results above apply for any r.

Fourth and finally, when $r \geq 1$ if the minimal fit-width solution is strictly better than any fit-height solution, then it corresponds uniquelly to a single point (see appendix for proof). Symmetrically for $r < 1$ the same applies for the fit-height solution.

We can now analyze when the algorithm we've developed fails. Translating its solution to points:

$$P_a = \begin{cases} (a_w = \lceil \sqrt{rn} \rceil, b_w = \lfloor \frac{a_w}{r} \rfloor)_0 & \text{if } n \leq a_w b_w \\ (a_h = a_w, b_h = \lceil \frac{a_w}{r} \rceil)_1 & \text{else} \end{cases}$$

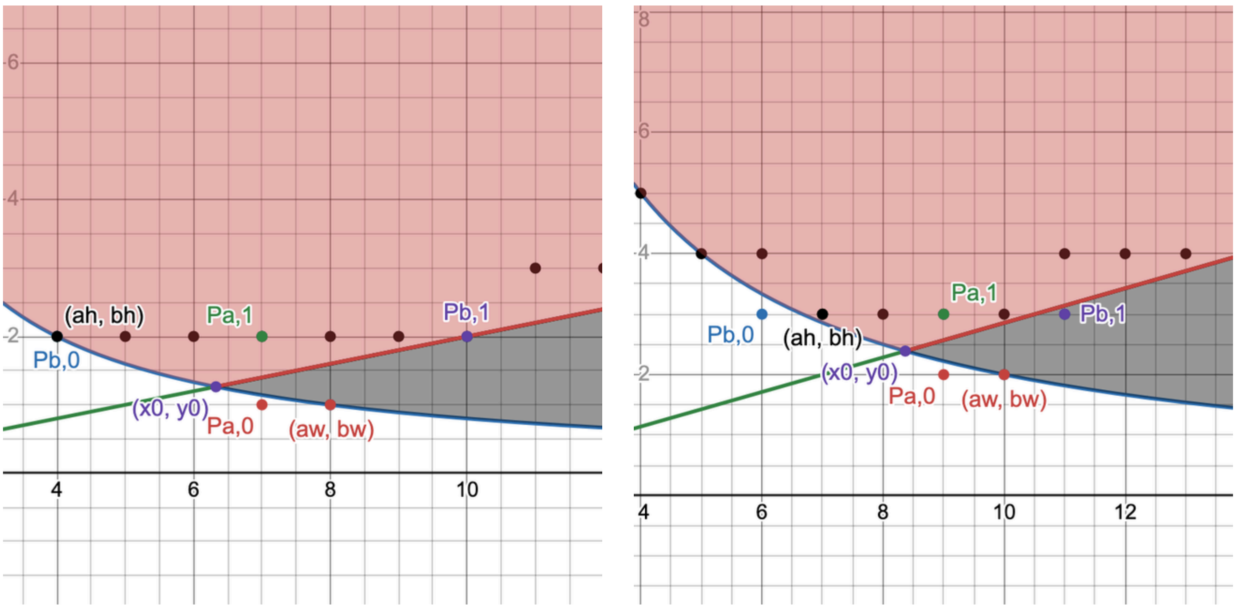
$$P_b = \begin{cases} (a_h = \lfloor r b_h \rfloor, b_h = \lceil \sqrt{\frac{n}{r}} \rceil)_0 & \text{if } n \leq a_h b_h \\ (a_w = \lceil r b_w \rceil, b_w = b_h)_1 & \text{else} \end{cases}$$

Recall that if a fit-width point has a solution greater than that of a fit-height point it will have less rows. Above we can see that all of $P_{a,1}, P_{b,0}, P_{b,1}$ have a number of rows greater than y_0: for P_b it is by construction, and for $P_{a,1}$:

$$\left\lceil \frac{a_w}{r} \right\rceil = \left\lceil \frac{\lceil \sqrt{rn} \rceil}{r} \right\rceil \geq \frac{\lceil \sqrt{rn} \rceil}{r} \geq \sqrt{\frac{n}{r}} = y_0$$

Moreover, for $r \geq 1$ there exists a fit-height solution with $b_h = \lceil y_0 \rceil$ with probability greater than $1 - \frac{1}{2r}$ (over 75% for $r \geq 2$) and fit-width solutions with $a_w = \lceil x_0 \rceil$ are highly unlikely – in fact, almost never likely for $r \geq 2$ (see appendix for proof). This means that the algorithm finds with very high guarantee the fit-height solution but rarely the fit-width one, if ever, even though the fit-width solution is most likely to be the optimal one.

Let's visualize this on the graphs of the two counterexamples given above ($r = 5, n = 8$ and $r = 3.5, n = 20$):



The initial fit-width point candidate $P_{a,0}$ is invalid in both cases (under the hyperbola), so the algorithm falls back to $P_{a,1}$, a fit-height point. In the first example, $P_{b,0}$ is already valid, so the fallback $P_{b,1}$ is not used; in the second, the fallback is used instead but even though it's a fit-width point its solution is equivalent to the fit-height solution, as they have the same number of rows.

Let's now build a new algorithm. This algorithm should find the minimal fit-width and fit height points and pick the solution with the greater side length. Putting (13) and (15) together, the solution we're looking for is:

$$
\begin{aligned}
a_w &= \min\left\{ a \in \mathbb{N} : a \geq \lceil \sqrt{rn} \rceil \wedge \left\lceil \frac{n}{a} \right\rceil \leq \left\lfloor \frac{a}{r} \right\rfloor \right\} \\
b_h &= \min\left\{ b \in \mathbb{N} : b \geq \left\lceil \sqrt{\frac{n}{r}} \right\rceil \wedge \left\lceil \frac{n}{b} \right\rceil \leq \lfloor rb \rfloor \right\} \\
s &= \max\left\{ \frac{w}{a_w}, \frac{h}{b_h} \right\}
\end{aligned}
$$

Let's walk through how we translate this to code. First, we are looking for minimum a_w and b_h greater than $\lceil x0 \rceil$ and $\lceil y0 \rceil$. We'll initialize variables `a` and `b` to those values, a_w and b_h can't be smaller. Then we need to ensure the second condition; for that, we just increment `a` and `b` in a loop until the condition holds. The loops execute while a_w and b_h are invalid points:

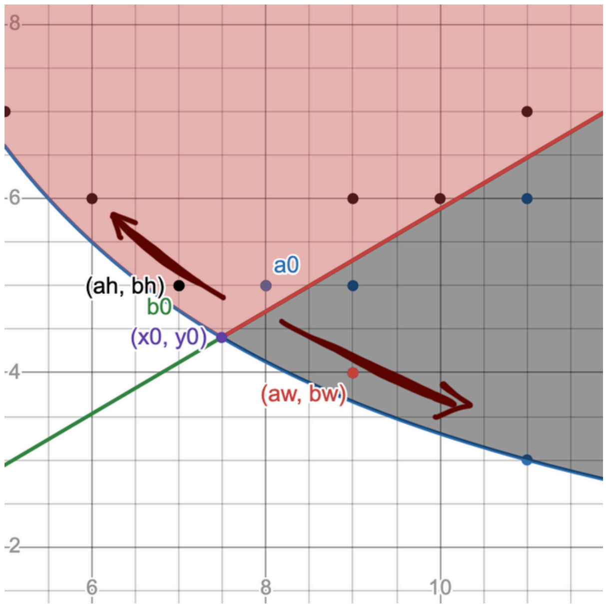
$$
\begin{aligned}
\left\lceil \frac{n}{a} \right\rceil > \left\lfloor \frac{a}{r} \right\rfloor &\iff a < r \left\lceil \frac{n}{a} \right\rceil \\
\left\lceil \frac{n}{b} \right\rceil > \lfloor rb \rfloor &\iff rb < \left\lceil \frac{n}{b} \right\rceil
\end{aligned} \tag{18}
$$

The implementation follows directly:

```
const r = w / h

let a = Math.ceil(Math.sqrt(r * n))
for (; a < r * Math.ceil(n / a); a++);

let b = Math.ceil(Math.sqrt(n / r))
for (; r * b < Math.ceil(n / b); b++);

return Math.max(w / a, h / b);
```

The algorithm always terminates: as a grows, $\left\lceil \frac{n}{a} \right\rceil$ decreases and $\left\lceil \frac{a}{r} \right\rceil$ increases, shrinking the interval until the ordering of its endpoints is swapped; likewise for b. The values at the end of each loop are minimal – we check every value one by one in ascending order.

Intuitively, if the loop condition holds there is no valid (a_w, b_w) or (a_h, b_h) fit-width, respectively fit-height point with the iteration's current a_w or b_h value. If we take $b_w = \left\lceil \frac{n}{a} \right\rceil$ and $a_h = \left\lceil \frac{n}{b} \right\rceil$ (the minimum values for a valid solution) we can imagine this algorithm as "walking away" from (x_0, y_0) alongside the hyperbola until we (inevitably) find ourselves in the fit-width/fit-height solution spaces:



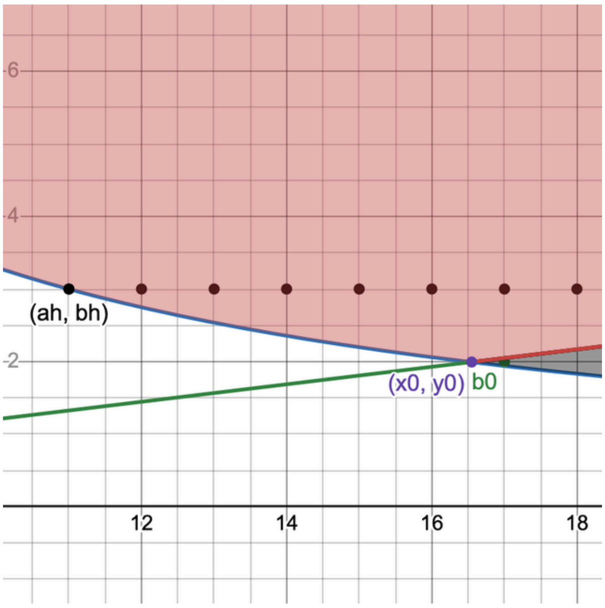
a_0 and b_0 are the starting points, right next to (x_0, y_0). The algorithm "walks down" to (a_w, b_w) and "up" to (a_h, b_h). In fact, in this example, b_0 is initialized directly to (a_h, b_h); a_0 must be incremented once.

We have convinced ourselves of correctness. To analyze its runtime complexity, let's denote $a_0 = \left\lceil \sqrt{rn} \right\rceil$ the starting value of a. Since the loop condition is $a < r \left\lceil \frac{n}{a} \right\rceil$, a will be incremented at most $\left\lceil r \left\lceil \frac{n}{a} \right\rceil - a_0 \right\rceil$ times. Since a increases $\left\lceil \frac{n}{a} \right\rceil \leq \left\lceil \frac{n}{a_0} \right\rceil$. We can now count iterations:

$$\begin{aligned}
\left\lceil r \left\lceil \frac{n}{a} \right\rceil - a_0 \right\rceil &< r \left\lceil \frac{n}{a_0} \right\rceil - a_0 + 1 \\
&< r \frac{n}{a_0} + r - a_0 + 1 \\
&= r \frac{n}{\left\lceil \sqrt{rn} \right\rceil} + r - \left\lceil \sqrt{rn} \right\rceil + 1 \\
&< \frac{rn}{\sqrt{rn}} + r - \sqrt{rn} + 1 \\
&= r + 1
\end{aligned} \tag{19}$$

This means at most $\lfloor r \rfloor + 1$ iterations and a runtime of $\mathcal{O}(r)$, dependent only on the aspect ratio of the grid. Binary search could also be used to reduce the runtime (see appendix).

To show that the bound is tight, consider this following example:



Here $n = 33, r = 8.3$. The fit-height starting point b_0 is in the fit-width solution space. Consider the iteration count of the fit-height loop of the algorithm – by the same reasoning it is $\lfloor \frac{1}{r} \rfloor + 1$. It's visible that in this example the loop will do exactly 1 iteration to reach (a_h, b_h). Since the algorithm is symmetric relative to r this goes as tightness proof for the first loop, too.

With a correct algorithm and good understang of why it works, we conclude here.

Appendix

Uniqueness of fit-width solution when $r \geq 1$

(9) implies that for b rows the minimal a such that (a, b) is a fit-width solution is:

$$a = \max\left\{ \left\lceil \frac{n}{b} \right\rceil, \lceil rb \rceil \right\}$$

Thus the minimal fit-width point is determined by:

$$a = \min_{b \in \mathbb{N}} \left\{ \max\left\{ \left\lceil \frac{n}{b} \right\rceil, \lceil rb \rceil \right\} \right\} \tag{*}$$

This makes a purely a function of b. This formula and a similar one for minimal fit-height solutions are used in the Desmos visualizer to plot minimal points for each a and b on the graph.

We now have all we need to begin the proof.

Assume towards a contradiction that there exist two $(a, b), (a, b')$ minimal fit-width points, $b < b'$, with $s_w > s_h$ for all fit-height points (a_h, b_h). Assume $r \geq 1$.

Since the points are minimal, it follows from $(*)$ that:

$$a = \max\left\{ \left\lceil \frac{n}{b} \right\rceil, \lceil rb \rceil \right\} = \max\left\{ \left\lceil \frac{n}{b'} \right\rceil, \lceil rb' \rceil \right\}$$

Without further deliberation, there are four situations we would have to tackle:

1. $a = \lceil rb \rceil = \lceil rb' \rceil$
2. $a = \left\lceil \frac{n}{b} \right\rceil = \left\lceil \frac{n}{b'} \right\rceil$
3. $a = \lceil rb \rceil = \left\lceil \frac{n}{b'} \right\rceil$
4. $a = \left\lceil \frac{n}{b} \right\rceil = \lceil rb' \rceil$

Intuitively, the first two shouldn't happen due to the monotonicity of rx and $\frac{n}{x}$. The third one also shouldn't happen, since (a, b) is around (x_0, y_0), where $b' > b$ would imply $\frac{n}{b'} < rb$. Only the fourth case seems legitimate. Let's confirm our intuition formally.

Disproving case 1 is easy:

$$\begin{aligned} b < b' \implies rb < rb' &\implies rb < \underbrace{rb + 1 \leq r(b+1)}_{\text{from } r \geq 1} \leq rb' \\ &\implies \lceil rb \rceil < \lceil rb \rceil + 1 \leq \lceil rb' \rceil \\ &\implies \lceil rb \rceil < \lceil rb' \rceil \;\; \text{contradiction} \end{aligned}$$

One would think disproving case 2 would be just as easy since it's still about monotonicity. Knowing that $r \geq 1$ helped tremendously above; doing the same for $\frac{n}{x}$ results in just $\lceil \frac{n}{b} \rceil \geq \lceil \frac{n}{b'} \rceil$. We need to get rid of the equality to actually have a contradiction, so let's find a sufficient condition for strict inequality:

$$
\begin{aligned}
b < b' \implies \frac{n}{b} > \frac{n}{b'} &\implies \frac{n}{b} > \frac{n}{b+1} \geq \frac{n}{b'} \\
\overbrace{\frac{n}{b} \geq \frac{n}{b+1} + 1}^{\text{condition candidate}} > \frac{n}{b+1} \geq \frac{n}{b'} &\implies \left\lceil \frac{n}{b} \right\rceil \geq \left\lceil \frac{n}{b+1} \right\rceil + 1 \geq \left\lceil \frac{n}{b'} \right\rceil \\
&\implies \left\lceil \frac{n}{b} \right\rceil > \left\lceil \frac{n}{b'} \right\rceil
\end{aligned}
$$

Perfect! When does this happen?

$$
\begin{aligned}
\frac{n}{b} \geq \frac{n}{b+1} + 1 &\iff \frac{n-b}{b} \geq \frac{n}{b+1} \\
&\iff b^2 + b - n \leq 0 \\
&\iff b \leq \frac{\sqrt{1+4n}-1}{2} \\
&\overset{\text{rm.}}{\iff} b \leq \left\lfloor \sqrt{\frac{1}{4}+n} - \frac{1}{2} \right\rfloor
\end{aligned}
\tag{**}
$$

Great. (I'm marking usage of residuated mapping properties with rm. since it's a subtle change easy to misuse). Is our b smaller than that?

$$
\begin{aligned}
a = \left\lceil \frac{n}{b'} \right\rceil \overset{(*)}{\implies} \left\lceil \frac{n}{b'} \right\rceil \geq \lceil rb' \rceil &\implies \left\lceil \frac{n}{b'} \right\rceil \geq rb' \geq b' \\
&\implies \frac{n}{b'} + 1 > b' \\
&\implies b'^2 - b' - n < 0 \\
&\implies b' < \frac{\sqrt{1+4n}+1}{2} \\
&\overset{\text{rm.}}{\implies} b' \leq \left\lfloor \sqrt{\frac{1}{4}+n} + \frac{1}{2} \right\rfloor
\end{aligned}
$$

$$
\begin{aligned}
b < b' &\implies b \leq \left\lfloor \sqrt{\frac{1}{4}+n} + \frac{1}{2} \right\rfloor - 1 \\
&\implies b \leq \left\lfloor \sqrt{\frac{1}{4}+n} - \frac{1}{2} \right\rfloor \\
&\overset{(**)}{\implies} \frac{n}{b} \geq \frac{n}{b+1} + 1 \\
&\implies \left\lceil \frac{n}{b} \right\rceil > \left\lceil \frac{n}{b'} \right\rceil \quad \text{contradiction}
\end{aligned}
$$

Yes, disproven! We can proceed with a counterproof for case 3, which is trivial:

$$\begin{aligned}
a = \left\lceil \frac{n}{b'} \right\rceil &\overset{(*)}{\implies} \left\lceil \frac{n}{b'} \right\rceil \geq \lceil rb' \rceil \\
&\implies \left\lceil \frac{n}{b} \right\rceil \geq \left\lceil \frac{n}{b'} \right\rceil \geq \overbrace{\lceil rb' \rceil > \lceil rb \rceil}^{\text{strict, see case 1}} \\
&\implies \left\lceil \frac{n}{b} \right\rceil > \lceil rb \rceil \\
a = \lceil rb \rceil &\overset{(*)}{\implies} \left\lceil \frac{n}{b} \right\rceil \leq \lceil rb \rceil \text{ contradiction}
\end{aligned}$$

Finally we are left with only case 4, $a = \lceil rb' \rceil = \left\lceil \frac{n}{b} \right\rceil$, the only one speculated to make sense.

To get an idea of what to prove, look at graph C again: there are two minimal fit-width points with equivalent solutions. Those points do not give the best solution: $a_w = 12 > rb_h = 3.9 \cdot 3$. Notice that one solution has $b_w = b_h$. Empirical observation on multiple parameters seems to always exhibit this, so let's prove that there is *always* an equivalent or better fit-height solution if $(a, b), (a, b')$ as chosen above are valid solutions.

Let's begin: since (a, b) and (a, b') are fit-width solutions, by (9) the following holds:

$$\begin{aligned}
n \leq ab \wedge a \geq rb \\
n \leq ab' \wedge a \geq rb'
\end{aligned} \tag{***}$$

Let $(a_h = \left\lceil \frac{n}{b'} \right\rceil, b_h = b')$. For (a_h, b_h) to be a valid fit-height point, (15) must hold true:

$$\left\lceil \frac{n}{b'} \right\rceil \leq \lfloor rb' \rfloor$$

Whether rb' is an integer or not changes the ceiling and floor result. Starting with $rb' \in \mathbb{N}$:

$$\begin{aligned}
rb' \in \mathbb{N} &\implies a = rb' \\
&\overset{(***)}{\implies} n \leq brb' \\
&\implies \frac{n}{b'} \leq rb < rb' \\
&\overset{\text{rm.}}{\implies} \left\lceil \frac{n}{b'} \right\rceil \leq \lfloor rb' \rfloor = rb'
\end{aligned}$$

It holds. For $rb' \notin \mathbb{N}$ let now $F = \lfloor rb' \rfloor$ and recall that $b' \geq b+1 > b$ by assumption. Then:

$$(i)\ \ a = \lceil rb' \rceil = F + 1$$

$$(ii)\ F = \lfloor rb' \rfloor \geq \underbrace{\lfloor r(b+1) \rfloor \geq b+1}_{r \geq 1} \implies \frac{F}{b+1} \geq 1$$

$$\begin{aligned}
\frac{n}{b'} \overset{(***)}{\leq} \frac{ab}{b'} &\overset{\text{(i)}}{=} \frac{(F+1)b}{b'} \\
&\leq \frac{(F+1)b}{b+1} \\
&= F\frac{b}{b+1} + \frac{b}{b+1} \\
&= F\frac{(b+1)-1}{b+1} + \frac{b}{b+1} \\
&= F - \frac{F}{b+1} + \underbrace{\frac{b}{b+1}}_{<1} \\
&\overset{\text{(ii)}}{\leq} F - 1 + 1 = F
\end{aligned}$$

$$\frac{n}{b'} \leq F \overset{\text{rm.}}{\iff} \left\lceil \frac{n}{b'} \right\rceil \leq \lfloor rb' \rfloor$$

With this we have shown that (15) holds for $(a_h = \lceil \frac{n}{b'} \rceil, b_h = b')$, meaning that (a_h, b_h) is a valid fit-height point. We assumed at the beginning that $s_w > s_h$. But by $(***)$:

$$a \geq rb' \iff \frac{1}{a} \leq \frac{1}{rb_h} \iff s_w \leq s_h \text{ contradiction}$$

All four cases contradict our assumptions. In conclusion, if (a, b) is a minimal fit-width point with $s_w > s_h$ for all valid fit-height (a_h, b_h) points, it must be unique.

Fun fact: the idea for disproving case 2 came from the intuition that since $n = \sqrt{n}^2$, dividing n by $b < b' < \sqrt{n}$ should mostly produce distinct results, as the value grows quicker and quicker as b approaches zero. $(**)$ implies exactly that:

$$b \leq \left\lfloor \sqrt{\frac{1}{4} + n} - \frac{1}{2} \right\rfloor \leq \left\lfloor \sqrt{(\frac{1}{2} + \sqrt{n})^2} - \frac{1}{2} \right\rfloor = \lfloor \sqrt{n} \rfloor$$

Position of solution points relative to (x_0, y_0)

We'll approach this subject by handling fit-height points where $r \geq 1$. Due to the symmetry of the problem relative to ratio, we can make more general conclusions afterwards.

$b = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point by (15) if and only if:

$$\left\lceil \frac{n}{b} \right\rceil \leq \lfloor rb \rfloor$$

A sufficient condition for this is:

$$\begin{aligned} rb - \frac{n}{b} \geq 1 &\implies rb^2 - n - b \geq 0 \\ &\implies b \geq \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} > \sqrt{\frac{n}{r}} \end{aligned}$$

If b is greater than this value it is guaranteed that there exists a fit-height point (a, b). There are more situations when this could happen not covered by this condition, for example when $\sqrt{\frac{n}{r}} \in \mathbb{N}$ or when $rb - \frac{n}{b} < 1 \land \frac{n}{b} \leq k \leq rb$ with $k \in \mathbb{N}$.

The condition above is useful since it is easy to *quanitfy*. We can use it to answer: for fixed $r \geq 1$ what's the probability for b to be a valid fit-height solution?

To do this, observe that by the properties of ceiling, $\sqrt{\frac{n}{r}} \leq b < \sqrt{\frac{n}{r}} + 1$. We can bound the expression above likewise, using $r \geq 1$:

$$\sqrt{\frac{n}{r}} < \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} < \frac{1}{2} + \sqrt{\left(\frac{1}{2} + \sqrt{\frac{n}{r}}\right)^2} = \sqrt{\frac{n}{r}} + 1$$

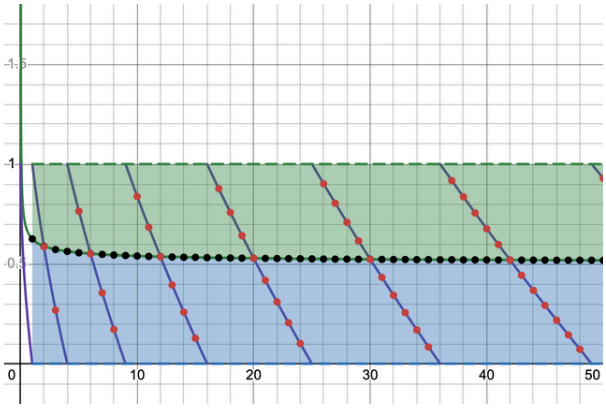
It follows that:

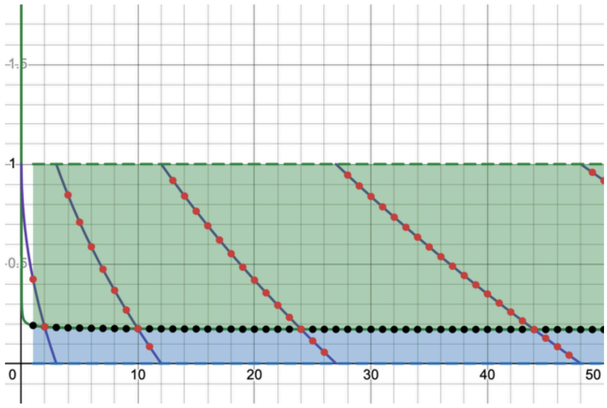
$$\sqrt{\frac{n}{r}} < \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} \leq b = \left\lceil \sqrt{\frac{n}{r}} \right\rceil < \sqrt{\frac{n}{r}} + 1$$

If we subtract $\sqrt{\frac{n}{r}}$, the difference will be always between 0 and 1:

$$0 < \underbrace{\frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} - \sqrt{\frac{n}{r}}}_{B_r(n)} \leq \underbrace{\left\lceil \sqrt{\frac{n}{r}} \right\rceil - \sqrt{\frac{n}{r}}}_{C_r(n)} < 1$$

Let's visualize this:





First graph depicts $r = 1$, the second $r = 3$. The red points represent $C_r(n)$, the black points $B_r(n)$. Whenever a red point is in the green area $C_r(n) \geq B_r(n)$, meaning $\left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point for n. The points where $C_r(n) = 0$ are ignored because that means that the corresponding b is an integer and thus a valid point.

The probability that b is a valid point up to a certain n of choice is the number of points in the green area divided by the total number of points:

$$P_r = \frac{|\{\, n \in \mathbb{N} : C(n) \geq B(n) \,\}|}{|\{\, n \in \mathbb{N} : C(n) > 0 \,\}|}$$

We can approximate by counting points up to a certain n. For example $P_{1,n=10} = \frac{4}{7} \approx 0.571$ and $P_{3,n=26} = \frac{21}{24} = 0.875$.

Counting points is difficult. Notice that on any interval where $C_r(x)$ is continuous for $x \in \mathbb{R}$, the points $(n, C_r(n))$ are distributed between the two areas proportionally with how much of the $C_r(x)$ segment is in either area. Since $0 < C_r(x) < 1$ covers the entire $(0, 1)$ space, the valid solutions area is the total area without the area of $B_r(x)$.

To compute this, let's simplify $B_r(n)$ first:

$$\begin{aligned}
B_r(n) &= \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} - \sqrt{\frac{n}{r}} \\
&= \frac{1}{2r} + \sqrt{\frac{n}{r}} \left(\sqrt{1 + \frac{1}{4rn}} - 1 \right) \\
&\leq \frac{1}{2r} + \sqrt{\frac{n}{r}} \cdot \frac{1}{8rn} \\
&= \frac{1}{2r} + \frac{1}{8r\sqrt{rn}}
\end{aligned}$$

The approximation $\sqrt{1+x} \leq 1 + \frac{x}{2}$ was used.

We can now observe the following properties of $B_r(x)$:

$$\lim_{x\to\infty} B_r(x) \leq \frac{1}{2r} \qquad \int_1^\infty B_r(x)dx = \infty$$

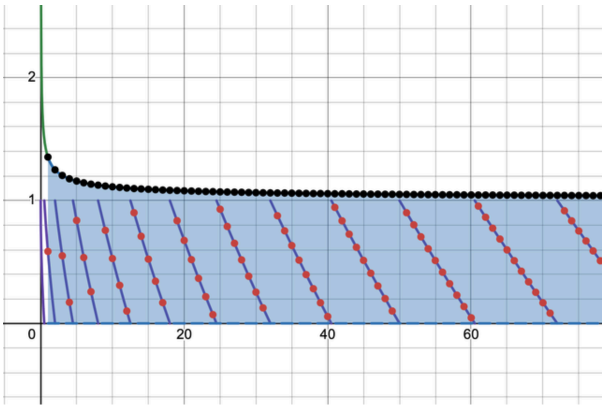
where the integral is the area under $B_r(x)$. We can now compute the probability:

$$P_r \geq 1 - \lim_{x\to\infty} \frac{\displaystyle\int_1^x B_r(n)dn}{\int_1^x 1} = 1 - \lim_{x\to\infty} B_r(x) \geq 1 - \frac{1}{2r}$$

Since we are in the $\frac{\infty}{\infty}$ indeterminate form l'Hôpital can be applied to avoid integration. P_r is greater than or equal to this result since we know there exist more successful cases beyond those included in this measurement.

Thus, for the examples above $P_1 \geq 0.5$ and $P_3 \geq 0.833$. This bound is more exact as n grows.

The first conclusion is that fit-height solutions will for the most part have $b_h = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$. We can derive even more interesting information by looking at the graph for $r < 1$:



Note that $r < 1$ means we've essentially flipped dimensions: the fit-height points correspond to the fit-width points of ratio $\frac{1}{r}$. For $r = \frac{1}{2}$ as depicted above, $P_{\frac{1}{2}} \geq 0$. How do we interpret this?

> For $r \geq 2$, no a = $\lceil \sqrt{rn} \rceil$ satisfies the sufficient condition above to fulfill the fit-width solution requirements.

Any valid a_w will be greater than $\lceil x_0 \rceil$ with very high certainty, especially when considering the growth rate of $\frac{1}{2r}$ as $r \to 0$. The second algorithm's runtime has further justification: as r grows the fit-width solutions are farther away from (x_0, y_0), so the first loop will iterate more; conversely, the initial fit-height solution candidate is more often valid, so the second loop iterates less.

The final conclusion is: for $r \geq 1$ it is very likely that $b_h = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point and almost never likely that $a_w = \lceil \sqrt{rn} \rceil$ is a valid fit-width point. Then mostly $b_w < b_h$, since with more columns the grid requires less rows to fit all squares. Therefore by (17) for $r \geq 1$ *the fit-width solution is in the majority of cases better than any fit-height solution*. Likewise for $r < 1$ but with fit-width and fit-height swapped.

Using binary search in the second algorithm

According to (15) the algorithm always finds a solution after $\lfloor r \rfloor + 1$ iterations, thus $a_0 \leq a_w < a_0 + r + 1$, where $a_0 = \lceil \sqrt{rn} \rceil$ is the starting value for a_w.

We can binary search over that interval using the same loop condition from (18). If the midpoint is not a valid a_w, we search in the upper half, since we're not yet in the solution space, otherwise we search in the lower half, since we want the minimal solution. Here is the algorithm for a_w:

```
const r = w / h

let left = Math.ceil(Math.sqrt(r * n))
let right = Math.ceil(left + r + 1)

while (left < right) {
    const mid = Math.floor(left + (right - left) / 2)
    if (mid < r * Math.ceil(n / mid)) {
        left = mid + 1
    } else {
        right = mid
    }
}

return w / left
```

Upper bound ceiled so interval is open. `left` stores a_w. This reduces the asymptotic runtime to $\mathcal{O}(\log r)$. Useful for very large ratios but very large ratios seem utterly useless.

$\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ are residuated mappings

This means that, for $\forall x \in \mathbb{R}, n \in \mathbb{N}$:

1. $n \leq x \iff n \leq \lfloor x \rfloor$
2. $x \leq n \iff \lceil x \rceil \leq n$
3. $n < x \iff n < \lceil x \rceil$
4. $x < n \iff \lfloor x \rfloor < n$

To prove the first, if $x \in \mathbb{N}$ then clearly $n \leq \lfloor x \rfloor = x$, else if $x \in \mathbb{R} \setminus \mathbb{N}$:

$$\begin{aligned} n \leq x \iff n < x &\iff n < \lfloor x \rfloor + \{x\} \\ &\iff n \leq \lfloor x \rfloor \vee (\lfloor x \rfloor < n < \lfloor x \rfloor + \{x\}) \\ &\overset{\{x\}<1}{\iff} n \leq \lfloor x \rfloor \vee (\lfloor x \rfloor < n < \lfloor x \rfloor + 1) \\ &\overset{n \in \mathbb{N}}{\iff} n \leq \lfloor x \rfloor \end{aligned}$$

To prove the second, if $x \in \mathbb{N}$ then clearly $x = \lceil x \rceil \leq n$, else if $x \in \mathbb{R} \setminus \mathbb{N}$:

$$\begin{aligned} x \leq n \iff x < n &\iff \lfloor x \rfloor + \{x\} < n \\ &\iff \{x\} < n - \lfloor x \rfloor \\ &\iff 1 \leq n - \lfloor x \rfloor \\ &\iff \lfloor x \rfloor + 1 \leq n \\ &\iff \lceil x \rceil \leq n \end{aligned}$$

The others are proven in a similar fashion. Read more about this on Wikipedia's "Equivalences" for floor and ceiling. These are standard properties but I insisted on writing them here as a "note to self" since I've confused myself and applied them wrong way too many times. I probably know that Wikipedia page now by heart.